

DRISHYAM

From Footage to Action

Edge Cases Document

Invalid Inputs, Failures & Recovery

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Confidential – For Development Use Only

Invalid / Malicious Input

- Malformed RTSP URL → 400 with clear message
- Extremely large JSON body → reject with 413
- SQL/NoSQL injection attempts → blocked by parameterization
- XSS in alert notes → escaped on output

Empty / Missing Data

- Org with zero cameras → empty state CTA
- Alert list empty → “All clear” message
- Clip not yet uploaded → “Processing...” with retry
- User has no site scope → see nothing or limited view

Duplicates & Concurrency

- Double-click acknowledge → idempotent action
- Two operators acknowledge same alert → first wins, second sees already-acked state
- Duplicate camera RTSP → warn or reject

Network & API Failure

- Backend unreachable from edge → local queue + retry with backoff
- WebSocket drop → reconnect + catch-up poll
- Push notification failure → retry + fallback to in-app only
- Partial form submit → preserve user input, show error

Auth & Authorization

- Expired access token → silent refresh or redirect to login
- User removed from org while session active → 403 on next request
- Role downgraded → UI and API both enforce new permissions immediately

AI / Pipeline Specific

- Model returns no detections for long period → health still OK if frames flowing
- Sudden burst of detections → rate-limit alerts per camera / cooldown
- Corrupt frame → skip and log
- Edge disk full → stop local clip write, still send metadata if possible, alert admin

Timeouts

- Long-running report → async job + notification when ready
- Camera test timeout → “Could not connect” after N seconds